Software Resources (also) For Preprocessing

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Intro

Many of the steps we have been discussing today are common to almost all software solutions availabe for tha analysis of metabolomics data:

1. Commercial suites
2. Open source solutions which can be run locally (e.g. `xcms`,`MzMine`,`MSDial`)
3. Freely available web services

XCMS Online

1 - Allow the use of xcms through a web interface

`xcms` online

2 - Allows to use xcms through Galaxy

The French metabolomics network also provide software for pathway analysis and visualization
`Metabohub` - Workflow-4-metabolomics

R based analysis

Linked to the Human Metabolome Database

MetaboAnalyst - statistical, functional and integrative analysis of metabolomics data
`Metaboanalyst`

Without R

Python Based

As part of the deNBI Center for integrative Bioinformatics, OpenMS offers an open-source software C++ library (+ python bindings) for LC/MS data management and analyses
`OpenMS`

In Galaxy and Knime

KNIME Analytics Platform is the open source software for creating data science
`knime`

The Phenomenal Project

PhenoMeNal (Phenome and Metabolome aNalysis) is a comprehensive and standardised e-infrastructure that supports the data processing and analysis pipelines for molecular phenotype data generated by metabolomics applications

`PhenoMeNal`

Human Friendly Open Solutions

MZmine

MZmine 3 is an open-source software for mass-spectrometry data processing, with the main focus on LC-MS data.
`MZmine3`

MS-Dial

MS-DIAL was launched as a universal program for untargeted metabolomics that supports multiple instruments (GC/MS, GC/MS/MS, LC/MS, and LC/MS/MS) and MS vendors (Agilent, Bruker, LECO, Sciex, Shimadzu, Thermo, and Waters)

`MS-Dial`